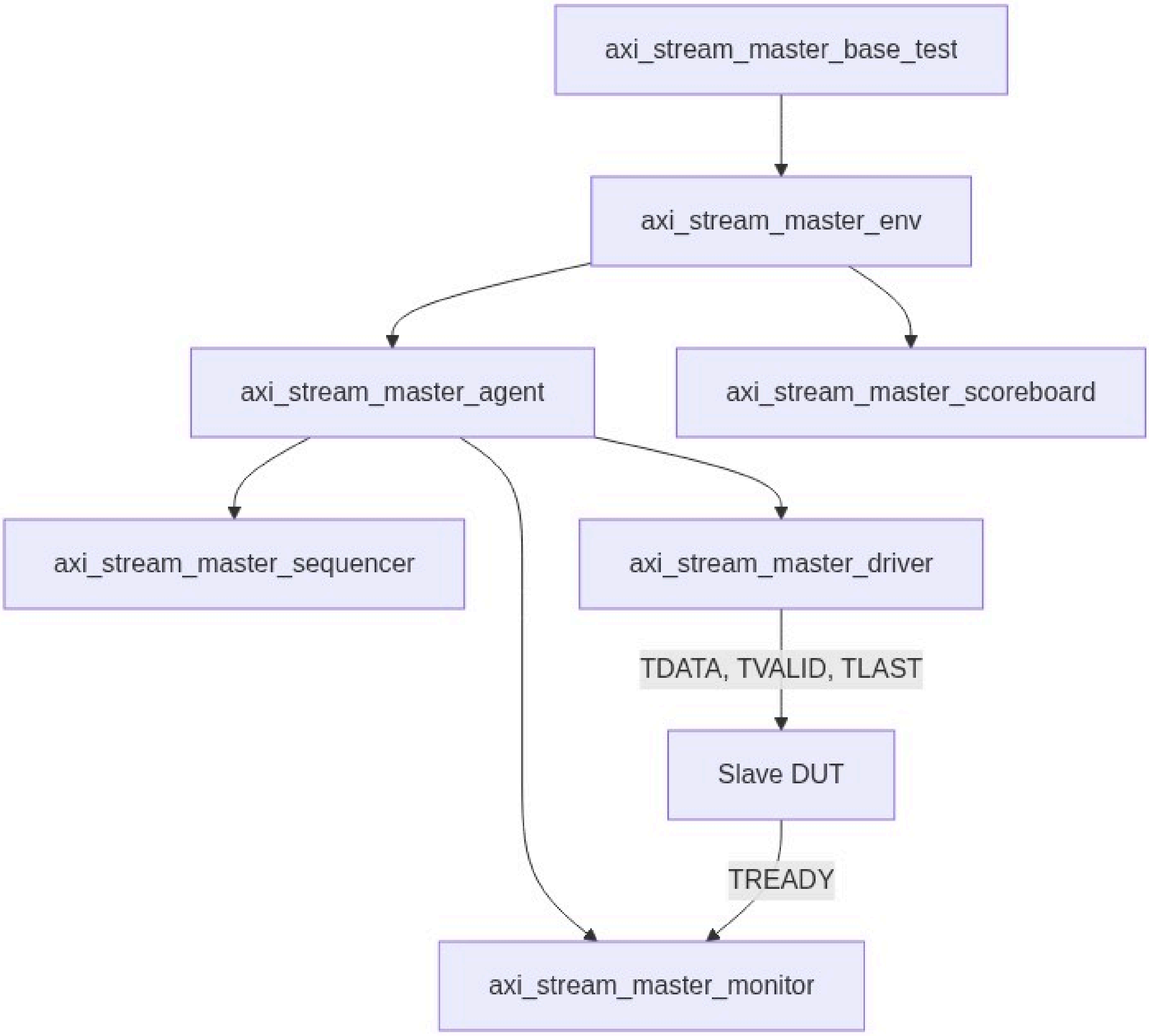
Verification Plan Report

Project: AXI-Stream Total Mastery

Version: 3.4

Verification Role: Master

1. UVM Environment Architecture



2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_MS_01	Handshake (Stability)	Master TDATA Stability	Master shall hold TDATA stable while observing TREADY=0.

Verification Plan Report

REQ_MS_02	Handshake (Independence)	TVALID/TREADY Flow	TVALID must be driven without waiting for Slave TREADY.
REQ_MS_03	Reset (Timing)	Reset Zero	Master must drive TVALID=0 during reset.
REQ_MS_04	Packet (Boundary)	TLAST Usage	TLAST must indicate the final cycle of every packet.

3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MS_01	Simple Handshake: Drive TVALID. Observe Slave DUT TREADY response.	REQ_MS_01
TC_MS_02	Back-pressure Stress: Observe 50 cycles of TREADY=0 from Slave DUT. Check stability.	REQ_MS_01
TC_MS_03	ZBT Packet: Drive TLAST=1 with all TKEEP=0. Observe Slave handshake.	REQ_MS_04

4. Functional Coverage Groups

Group Name	Description
cg_handshake	Cross of Master TVALID x Observed Slave TREADY.

5. Interface Signals

Signal Name	Direction	Width	Description
TDATA	Output	TD_W	Master data.
TVALID	Output	1	Master valid.
TREADY	Input	1	Slave ready (Observed).

6. Verification Environment Structure

`verif/axis_master_v3_4/`

7. SystemVerilog Implementation

axi_stream_master_seq_item.sv

```
COMPONENT: axi_stream_master_seq_item
EXTENDS: uvm_sequence_item
MEMBERS:
  - [rand logic [TD_W-1:0]] tdata
  - [rand logic] tlast
  - [rand int] inter_beat_delay
```

```
LOGIC:
  - Constraint: delay between 0 and 10 cycles.
```

axi_stream_master_driver.sv

```
COMPONENT: axi_stream_master_driver
EXTENDS: uvm_driver
LOGIC:
  - Action: Drive TVALID=1 and TDATA.
  - Wait: Observe Slave DUT TREADY=1 before proceeding.
```

axi_stream_master_monitor.sv

```
COMPONENT: axi_stream_master_monitor
EXTENDS: uvm_monitor
LOGIC:
  - Sample: If TVALID && TREADY, capture item.
  - Check: Verify TDATA stability during TREADY=0 stall.
```

axi_stream_master_agent.sv

```
COMPONENT: axi_stream_master_agent
EXTENDS: uvm_agent
MEMBERS: Driver, Monitor, Sequencer
LOGIC:
  - Connection: Connect Driver's seq_item_port to Sequencer.
```

axi_stream_master_agent_config.sv

```
COMPONENT: axi_stream_master_agent_config
EXTENDS: uvm_object
MEMBERS: Virtual Interface handle, is_active flag.
```

axi_stream_master_scoreboard.sv

```
COMPONENT: axi_stream_master_scoreboard
EXTENDS: uvm_scoreboard
MEMBERS: Analysis FIFO, Reference queue
LOGIC:
  - Comparison: Match monitored Master output against expected sequence.
```

axi_stream_master_env.sv

```
COMPONENT: axi_stream_master_env
EXTENDS: uvm_env
MEMBERS: Master Agent, Scoreboard
LOGIC:
  - Connect: Monitor analysis port to Scoreboard.
```

Verification Plan Report

axi_stream_master_env_config.sv

```
COMPONENT: axi_stream_master_env_config
EXTENDS: uvm_object
MEMBERS: Agent Config handles, scoreboard enable switch.
```

axi_stream_master_base_seq.sv

```
COMPONENT: axi_stream_master_base_seq
EXTENDS: uvm_sequence
LOGIC:
  - Loop: 10 times: start_item(req), randomize, finish_item(req).
```

axi_stream_master_base_test.sv

```
COMPONENT: axi_stream_master_base_test
EXTENDS: uvm_test
CONFIG_FLOW:
  - Initialize Config Object.
  - Set Virtual Interface in ConfigDB.
  - Start Base Sequence on Sequencer.
```

axi_stream_if.sv

```
COMPONENT: axi_stream_if
MEMBERS:
  - logic ACLK, ARESETn
  - logic [TD_W-1:0] TDATA
  - logic TVALID, TREADY, TLAST
LOGIC:
  - Assertions: Protocol stability and reset-low TVALID checks.
```